P1

(1) $m\vec{a} = q\vec{E}(t) - k\vec{r} - m\gamma\vec{v}$, $k = m\omega_0^2$

$$\vec{v}(t) = \frac{d\vec{r}}{dt} \quad , \vec{a}(t) = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

Fourier $\vec{r}(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t} \vec{r}_\omega$

$$\vec{v}(t) = \frac{d}{dt}\int \frac{d\omega}{2\pi} e^{-i\omega t}\vec{r}_\omega = \int \frac{d\omega}{2\pi} -i\omega e^{-i\omega t}\vec{r}_\omega$$

$$\vec{a}(t) = \int \frac{d\omega}{2\pi}(-i\omega)^2 e^{-i\omega t}\vec{r}_\omega$$

$$\vec{E}(t) = \int \frac{d\omega}{2\pi} e^{-i\omega t}\vec{E}_\omega$$

Eq. 1 becomes

$$\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t}(-m\omega^2)\vec{r}_\omega = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t}\left\{ q\vec{E}_\omega - k\vec{r}_\omega - m\gamma(-i\omega)\vec{r}_\omega \right\}$$

$$-m\omega^2\vec{r}_\omega = q\vec{E}_\omega - k\vec{r}_\omega + im\gamma\omega\vec{r}_\omega$$

$$(m\omega^2 - k + im\gamma\omega)\vec{r}_\omega = -q\vec{E}_\omega$$

$$\vec{r}_\omega = (-q)\frac{\vec{E}_\omega}{m(\omega^2-\omega_0^2) + im\gamma\omega}$$

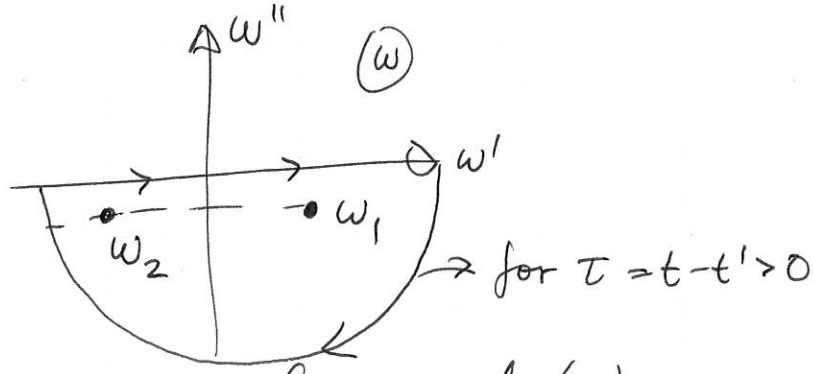
$$\vec{r}(t) = \int \frac{d\omega}{2\pi} e^{-i\omega t}\vec{r}_\omega = -q\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{\vec{E}_\omega e^{-i\omega t}}{m(\omega^2-\omega_0^2) + im\gamma\omega}$$

$$= -\frac{q}{m}\int \frac{d\omega}{2\pi} \frac{\vec{E}_\omega e^{-i\omega t}}{\omega^2 + i\gamma\omega - \omega_0^2} =$$

$$= -\frac{q}{m}\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{\vec{E}_\omega e^{-i\omega t}}{(\omega-\omega_1)(\omega-\omega_2)}$$

where $\omega_{1,2}$ are the roots of

$$\omega^2 + i\gamma\omega - \omega_0^2 = 0 \Rightarrow \omega_{1,2} = -i\frac{\gamma}{2} \pm \sqrt{\omega_0^2 - \left(\frac{\gamma}{2}\right)^2}$$



Integration is along real (ω') axis.
Two poles are in $\omega'' = \mathrm{Im}(\omega) < 0$ half-plane.

Inverse transform $\vec{E}_\omega = \int_{-\infty}^{\infty} dt'\, e^{i\omega t'}\, \vec{E}(t')$

$$\vec{r}(t) = -\frac{q}{m}\int_{-\infty}^{\infty} dt'\, \vec{E}(t') \int_{-\infty}^{\infty} \frac{d\omega}{2\pi}\, \frac{e^{-i\omega(t-t')}}{(\omega-\omega_1)(\omega-\omega_2)}$$

For $\tau = t - t' > 0$ we can close contour of integration in lower half-plane of ω because $e^{-i\omega\tau} = e^{-i\tau(\omega' + i\omega'')}$
$= e^{-i\tau\omega' + \tau\omega''}$ converges for $\omega'' < 0$.

For $\tau = t - t' < 0$ we need to close it in upper half-plane where $e^{-i\omega\tau} = e^{-i\tau(\omega' + i\omega'')} = e^{-i\tau\omega' - \tau\omega''}$ converges for $\omega'' > 0$.

But there are no poles there!

$\underline{\tau > 0}$ $I = \int \frac{d\omega}{2\pi} \frac{e^{-i\omega\tau}}{(\omega-\omega_1)(\omega-\omega_2)} = -\frac{2\pi i}{2\pi}\left(\frac{e^{-i\omega_1\tau}}{\omega_1-\omega_2} + \frac{e^{-i\omega_2\tau}}{\omega_2-\omega_1}\right) =$

$$= (-i)\left(e^{-\frac{\gamma\tau}{2} - \sqrt{\omega_0^2 - \gamma^2/4}\,\tau} - e^{-\gamma\tau/2 + \sqrt{\omega_0^2 - \gamma^2/4}}\right)\frac{1}{2\sqrt{\omega_0^2 - \gamma^2/4}}$$

$$I = (-i)\, e^{-\frac{\gamma\tau}{2}} \frac{-2i \sin\sqrt{\omega_0^2 - \frac{\gamma^2}{4}}\,\tau}{2\sqrt{\omega_0^2 - \frac{\gamma^2}{4}}} \xrightarrow[\gamma \to 0]{} -\frac{\sin(\omega_0 \tau)}{\omega_0}$$

for $\tau > 0$

$I = 0$ for $\tau < 0$.

$$\vec{r}(t) = +\frac{q}{m} \int_{-\infty}^{t} dt'\, \vec{E}(t') \frac{\sin(\omega_0 \tau)}{\omega_0}$$

$\tau \equiv t - t' > 0 \Rightarrow \underline{t' < t}$ ← upper limit of integration

Thus, small $\gamma \to 0$ has provided us with physically motivated choice of selecting poles for the ω-integration.

(P2)

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\Delta J}{\rho}\,\delta(x)\,\delta(t)$$

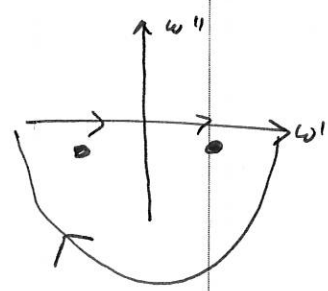
$$u(t,x) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int_{-\infty}^{\infty} \frac{dk}{2\pi}\, e^{ikx - i\omega t}\, u(\omega,k)$$

$$\delta(x) = \int \frac{dk}{2\pi}\, e^{ikx} \quad , \delta(t) = \int \frac{d\omega}{2\pi}\, e^{-i\omega t}$$

$$\Rightarrow \left(c^2 k^2 - \omega^2\right) u(\omega,k) = \frac{\Delta J}{\rho}$$

$$u(t,x) = \int \frac{d\omega\, dk}{(2\pi)^2}\, e^{ikx - i\omega t} \frac{(-\Delta J/\rho)}{(\omega^2 - c^2 k^2)} =$$

$$= \int \frac{dk}{2\pi} \int \frac{d\omega}{2\pi} \frac{e^{ikx}(-\Delta J/\rho)\, e^{-i\omega t}}{(\omega - ck + i\eta)(\omega + ck + i\eta)}$$



$$= -\frac{\Delta J}{\rho} \int_{-\infty}^{\infty} \frac{dk}{2\pi}\, e^{ikx}\, \frac{-2\pi i}{2\pi} \left(\frac{e^{-ickt - \eta t}}{2ck} + \right.$$

for $t>0$

$$\left. + \frac{e^{ickt - \eta t}}{-2ck} \right) = i\,\frac{\Delta J}{\rho}\, \underbrace{\theta(t)}_{\text{step function}} \int_{-\infty}^{\infty} \frac{dk}{2\pi}\, \frac{e^{ik(x-ct)} - e^{ik(x+ct)}}{2ck}$$

$$\int_{-\infty}^{\infty} \frac{dk}{2\pi}\, \frac{e^{ik\xi}}{k} = \int_0^{\infty} \frac{dk}{2\pi}\, \frac{e^{ik\xi}}{k} + \underbrace{\int_{-\infty}^{0} \frac{dk}{2\pi}\, \frac{e^{ik\xi}}{k}}_{-\int_0^{\infty} \frac{dk'}{2\pi} \frac{e^{-ik'\xi}}{k'}} =$$

$$= \int_0^{\infty} \frac{dk}{2\pi}\, \frac{\left(e^{ik\xi} - e^{-ik\xi}\right)}{k}$$

$$u(t,x) = i\,\frac{\Delta J}{\rho}\,\theta(t)\,\frac{2i}{4\pi c}\int_0^\infty \frac{dk}{k}\left(\sin k(x-ct) - \sin k(x+ct)\right)$$

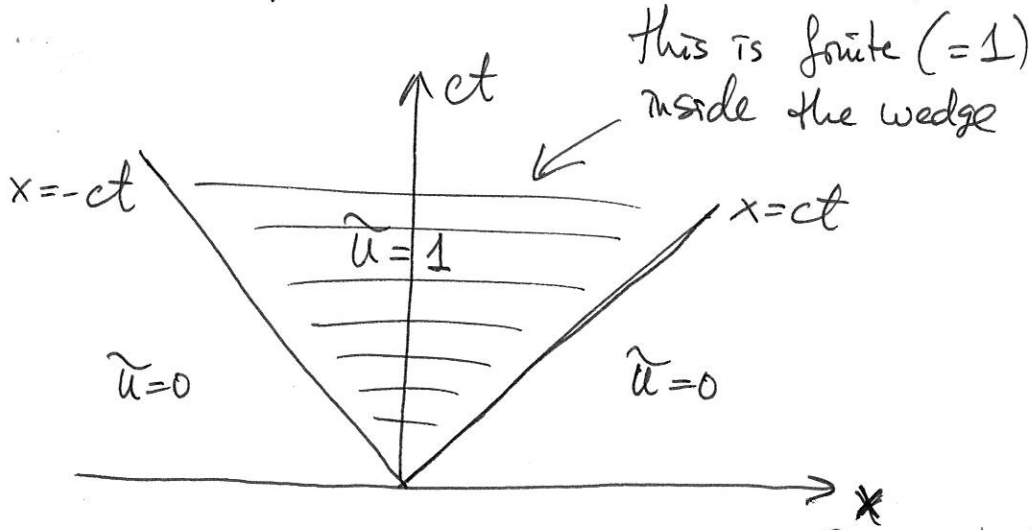
But $\displaystyle\int_0^\infty dk\,\frac{\sin(ak)}{k} = \frac{1}{2}\int_{-\infty}^{\infty} dk\,\frac{\sin(ak)}{k} = \frac{\pi}{2}\,\mathrm{sign}(a)$

$$u(t,x) = -\frac{\Delta J\,\theta(t)}{\rho\cdot 2\pi c}\,\frac{\pi}{2}\left(\mathrm{sign}(x-ct) - \mathrm{sign}(x+ct)\right)$$

$$= \frac{\Delta J}{4c\rho}\,\theta(t)\left[\mathrm{sign}(x+ct) - \mathrm{sign}(x-ct)\right].$$

Use $\mathrm{sign}(a) = 2\theta(a) - 1 = \begin{cases} 1, & a>0 \\ -1, & a<0 \end{cases}$

$$\Rightarrow u(t,x) = \frac{\Delta J}{2c\rho}\,\theta(t)\left(\theta(x+ct) - \theta(x-ct)\right)$$



The function $\tilde{u}(t,x) = \dfrac{2c\rho}{\Delta J}\,u(t,x) = \begin{cases} 1, & |x|<ct \\ 0, & \text{otherwise} \end{cases}$